

Handoff Contracts, Input Flexibility, and Context Pruning

Method (300w) + Stage 1, 3, 4 Redefinition (800+w) + Pruning Redesign | MECE

Part 1: The Method — How Stages Connect

Every stage is a **contract**: it declares what it requires to start (input contract), what it guarantees when finished (output contract), and what it passes forward (handoff envelope). A handoff envelope is the specific slice of an artifact the next stage needs — never the full artifact unless justified.

Input acquisition follows a fixed protocol at pipeline start. Stage 0 prompts the user:

- 1. **Text input (mandatory, never skippable)**: Describe the idea, problem, or feature. Minimum 2 sentences.
- 2. **Optional enrichment (user chooses which, if any)**: Screenshots, URLs, Files, Voice/transcription. Presented as checkboxes: 'Do you have any of these?' User can skip all optional. Text is the seed.

Cross-stage feedback uses a **flag-and-patch** mechanism. If a downstream stage finds upstream data missing, it emits a `DEPENDENCY_GAP` flag with: (a) what's missing, (b) which upstream section should contain it, (c) whether it blocks or degrades quality. The orchestrator routes the gap back upstream for patching.

Quality gates at each transition validate that the output contract is met before the handoff envelope is assembled. Validation is structural (sections present, word counts, tables exist) and substantive (evidence density, citation tiers). A stage cannot hand off until validation passes.

Part 2: Stage Redefinitions

Stage 1: Research — Redefined Input Contract

Problem: Stage 1 assumes input is a topic string. No mechanism for screenshots, URLs, or files. The user's idea gets flattened to a keyword.

Input Field	Req?	Format	How Stage 1 Uses It
idea_text	YES	Free text (min 2 sentences)	Seed's decision-to-inform framing. Researcher extracts: domain, competitors, target user, problem.
screenshots[]	No	PNG/JPG	Analyzed for current UX state, feature gaps. Each annotated. Becomes Tier 1 evidence.
urls[]	No	Web URLs	Fetches as starting sources. Classified by evidence tier. Accelerates competitor analysis.
files[]	No	PDF/DOCX/MD/PPTX	Parsed for existing research, prior PRDs, customer data. Cited as Tier 1-2 if primary.
constraints	No	Free text	Timeline, budget, team, stack, compliance. Passed to ALL downstream stages.

Output Contract — what Stage 1 guarantees:

Section	Required?	Minimum Quality Bar
Decision to Inform	Yes	Single sentence, links to idea_text
Executive Summary	Yes	200-300 words, conclusion-first
Own Product Gap Analysis	Yes	>= 1,500 words, names specific capabilities
Primary Competitor	Yes	>= 800 words, evolution timeline table
Secondary Competitors	Yes	>= 400 words each, timeline per competitor
Market Data Table	Yes	>= 8 rows with Tier column

Section	Required?	Minimum Quality Bar
Pricing Comparison	Yes	Entry for every named competitor
TAM Calculation	Yes	Bottoms-up math (not just a number)
Customer Voice	Yes	>= 3 direct quotes with attribution
Key Takeaways	Yes	5-7 actionable items
Evidence Tier Distribution	Yes	>= 40% Tier 1-2

Handoff Envelopes from Stage 1:

To Stage 2: Full research + user_inputs.md (original idea text, constraints, files/URLs)

To Stage 3: Research Summary (Exec Summary + Takeaways + Market Data + TAM + Pricing, ~1,500w) + pricing_table + customer_voice extracts

To Stage 4: Competitive Context (Exec Summary + Primary Competitor Thesis + Implication, ~600w)

Stage 3: Gandalf — Redefined with Cross-Stage Feedback

Problem: Gandalf receives blob input. If PRD is missing a TAM section, Q1 fails — but that's an input gap, not a strategy gap. No mechanism to route gaps upstream.

Input Contract:

Input	Source	How Gandalf Uses It
prd-v[N].md (full)	Stage 2	Primary artifact under evaluation
research_summary	Stage 1 (pruned)	Cross-ref PRD claims for Q1 (TAM), Q2 (Why Now), Q5 (Moat), Q8 (Cannib.)
user_constraints	Stage 0	Adjusts scoring — e.g. if "no pricing yet," Q10 bar is lower
research_pricing_table	Stage 1 (extract)	Feeds Q10 directly. If missing, Gandalf knows it is a pipeline gap.
research_customer_voice	Stage 1 (extract)	Feeds Q3. Checks if PRD scenarios trace to real quotes.

Pre-Evaluation Structural Check (NEW): Before scoring, Gandalf verifies each of its 10 questions has a corresponding PRD section at minimum depth:

Question	Required PRD Section	Minimum Check
Q1: TAM	Quantitative data or FAQs	TAM number + math present
Q2: Why Now	Exec Summary or Problem Depth	Temporal trigger identified
Q3: Problem Depth	Persona + Problem Depth	>= 3 named scenarios
Q4: North Star	Success Metrics	Metric + rejected alternatives
Q5: Moat	Competitive Differentiation	>= 1 structural advantage
Q6: Scope	Scope Boundary Table	Table with rationale column
Q7: Tech Feasibility	Solution or FAQs	Technical approach described
Q8: Cannibalization	Risks or FAQs	Existing services named
Q9: Failure Mode	Risks	>= 1 falsifiable scenario
Q10: Pricing	FAQs or Business	Revenue path described

DEPENDENCY_GAP Mechanism: If structural check finds a section missing, Gandalf emits a flag: question ID, what's missing, which PRD section should contain it, severity (BLOCKING vs WARNING). Orchestrator routes back to PRD Writer for patching. Gandalf continues scoring non-blocked questions in parallel.

New Output Sections:

Section	Content	Consumed By
Detailed Scores (existing)	10-row rubric + evidence table	Orchestrator, Human
Dependency Gaps Emitted (NEW)	DEPENDENCY_GAP flags sent upstream	Orchestrator, PRD Writer
Evidence Map (NEW)	Per passing Q: PRD section + research source the stage	Stage 4 Designer
Design-Relevant Flags (NEW)	Q6 scope cuts + Q7 tech constraints + Q9 failure modes	Stage 4 Designer, Stage 5

Stage 4: Designer — Redefined with Structured Extraction

Problem: Designer receives blob PRD + Gandalf evaluation. Re-reads entire PRD cherry-picking what it thinks matters. No structured extraction of personas, JTBD, scope, or validated claims — leading to inconsistent quality.

Input Contract — each field extracted deterministically:

Input	Source	Extracted From	How Designer Uses It
primary_persona	Stage 2	PRD Primary Persona	Seeds Phase 1 first-principles (task, needs, expertise)
secondary_persona	Stage 2	PRD Secondary Persona	Expert-vs-novice design (Checklist #9)
jtbd_ranked	Stage 2	PRD JTBD section	Maps to info priority hierarchy (JTBD #1 = what user sees first)
solution_capabilities	Stage 2	PRD Solution Proposal	Each capability becomes a UI element to design
scope_boundary_table	Stage 2	PRD Scope Boundary	Designer ONLY designs v1 column, omits rest
success_metrics	Stage 2	PRD Success Metrics	North Star = most prominent data point in UI
gandalf_evidence_map	Stage 3	Gandalf Evidence Map	Knows which claims validated (design confidently) vs flagged
gandalf_design_flags	Stage 3	Gandalf Q6+Q7+Q9	Scope cuts, failure modes, tech constraints
competitive_context	Stage 1	Research Exec + Primary Pattern	Pattern selection (match vs differentiate)
user_screenshots	Stage 0	User-provided images	Baseline for improvement annotations

Pre-Design Structural Check: If primary_persona < 150w or jtbd_ranked < 3 items or scope_boundary_table missing, Designer emits DEPENDENCY_GAP to PRD Writer and proceeds with degraded defaults.

Output Contract — new sections added from GitHub UX research:

Section	Required	Minimum Quality Bar
Primary User Task	Yes	Single sentence from JTBD #1
Information Priority	Yes	>= 3 priorities linked to JTBD
Critical User Journey	Yes	5-7 steps, 4 columns each (Action/Response/Data/Component)
Alternatives Considered	Yes	>= 2 per major decision, with Why Chosen Wins
Cloudscape Component Mapping	Yes	Every element mapped to named component
Design Checklist	Yes	10/10 scored, minimum 7/10 >= 3
Error & Edge States	Yes	>= 4 states (empty, error, loading, partial)
Stickiness Design	Yes	>= 2 habit loops
Prototype Handoff Notes	Yes	Interactions + mock data + breakpoints
Nielsen Heuristic Audit (NEW)	Yes	All 10 heuristics scored 1-5
Interaction State Spec (NEW)	Yes	Hover/focus/active/disabled per component
Keyboard Navigation Spec (NEW)	Yes	Tab order, focus mgmt, power-user shortcuts

Part 3: Context Pruning Redesign

Current problems: 'summary' is undefined (how long? what sections?). No extracted fields — just blobs. No user context after Stage 0. No feedback mechanism.

Redesigned Handoff Flow

Each arrow is a named, validated envelope. Extraction is deterministic (section heading match + word count), not LLM summarization.

From	Envelope	To	Max Size	Extraction Rule
Stage 0	idea_text + constraints	ALL stages	~500w	Persists unchanged
Stage 0	screenshots[]	S1, S4, S5	N/A	Images passed by reference
Stage 1	research (FULL)	Stage 2 only	5-7.5Kw	Only stage that gets full research
Stage 1	research_summary	S3, S6	1,500w	Exec Summary + Takeaways + Market Data + TAM + Pricing
Stage 1	research_competitive_context	Stage 4	600w	Exec Summary + Primary Competitor Thesis + Implication
Stage 1	research_pricing_table	Stage 3	table	Pricing Comparison table only
Stage 1	research_customer_voice	Stage 3	section	Customer Voice section only
Stage 2	prd (FULL)	S3, S4	5.5-8Kw	Gandalf evaluates full; Designer needs full
Stage 2	prd_executive_summary	S5, S6	250w	Executive Summary section
Stage 2	prd_personas	Stage 4	extract	Primary + Secondary Persona sections
Stage 2	prd_jtbd_ranked	Stage 4	extract	JTBD section with rankings
Stage 2	prd_scope_boundary	Stage 4	table	Scope Boundary table
Stage 2	prd_success_metrics	S4, S6	section	Success Metrics section
Stage 2	prd_risks	Stage 6	section	Risks + Open Questions section
Stage 3	gandalf (FULL)	Stage 4	eval doc	Designer needs evidence map + flags
Stage 3	gandalf_design_flags	S4, S5	400w	Q6 scope + Q7 tech + Q9 failure
Stage 3	gandalf_score_summary	Stage 6	200w	Verdict + score table
Stage 4	design-spec (FULL)	Stage 5	design doc	Prototype Builder needs everything
Stage 4	design_component_mapping	Stage 6	table	Component table
Stage 4	design_interaction_list	Stage 6	list	Critical interactions from handoff
Stage 4	design_error_states	Stage 6	table	Error/edge state table
Stage 5	prototype path + notes	Stage 6	path+notes	File path + validation results

Cross-Stage Feedback Protocol

1. Downstream stage emits DEPENDENCY_GAP (what's missing + which upstream section + severity)
2. Orchestrator logs in pipeline-state.md under Dependency Gaps section
3. Orchestrator invokes upstream skill with PATCH instruction
4. Upstream patches, emits v[N+1] artifact
5. Orchestrator re-extracts affected handoff envelope
6. Downstream re-scores only affected question/section

This happens in parallel — downstream continues non-blocked work while upstream patches.

Token Budget per Stage

Stage	Max Input	Breakdown
1 (Research)	~3K tokens	idea_text + urls + constraints + file summaries
2 (PRD Writer)	~12K tokens	research (full 5-7.5Kw) + user_inputs
3 (Gandalf)	~15K tokens	prd (full 5.5-8Kw) + research_summary (1.5Kw) + tables
4 (Designer)	~18K tokens	prd (full) + gandalf (full) + competitive context (600w)
5 (Prototype)	~10K tokens	design-spec (full) + prd summary (200w) + flags
6 (Launch)	~8K tokens	All extracted envelopes (no full artifacts except proto-notes)